

Semantic Caching: 30x Faster RAG Response Times

Leveraging intelligent query caching to dramatically reduce latency in production AI systems

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Tech Stack

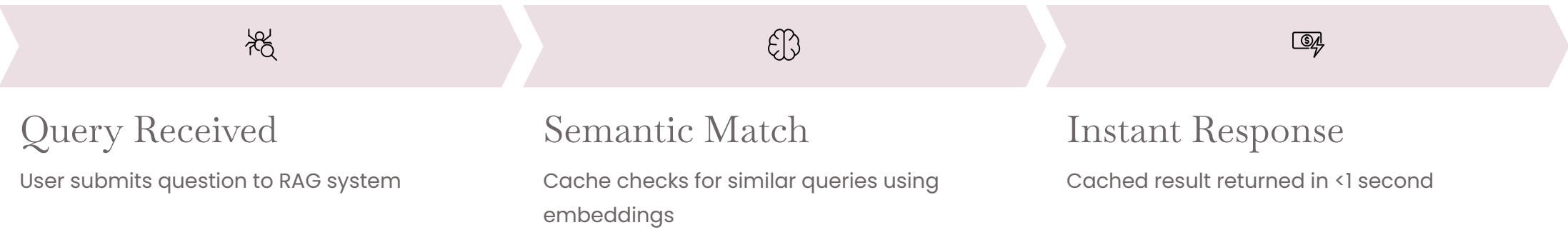
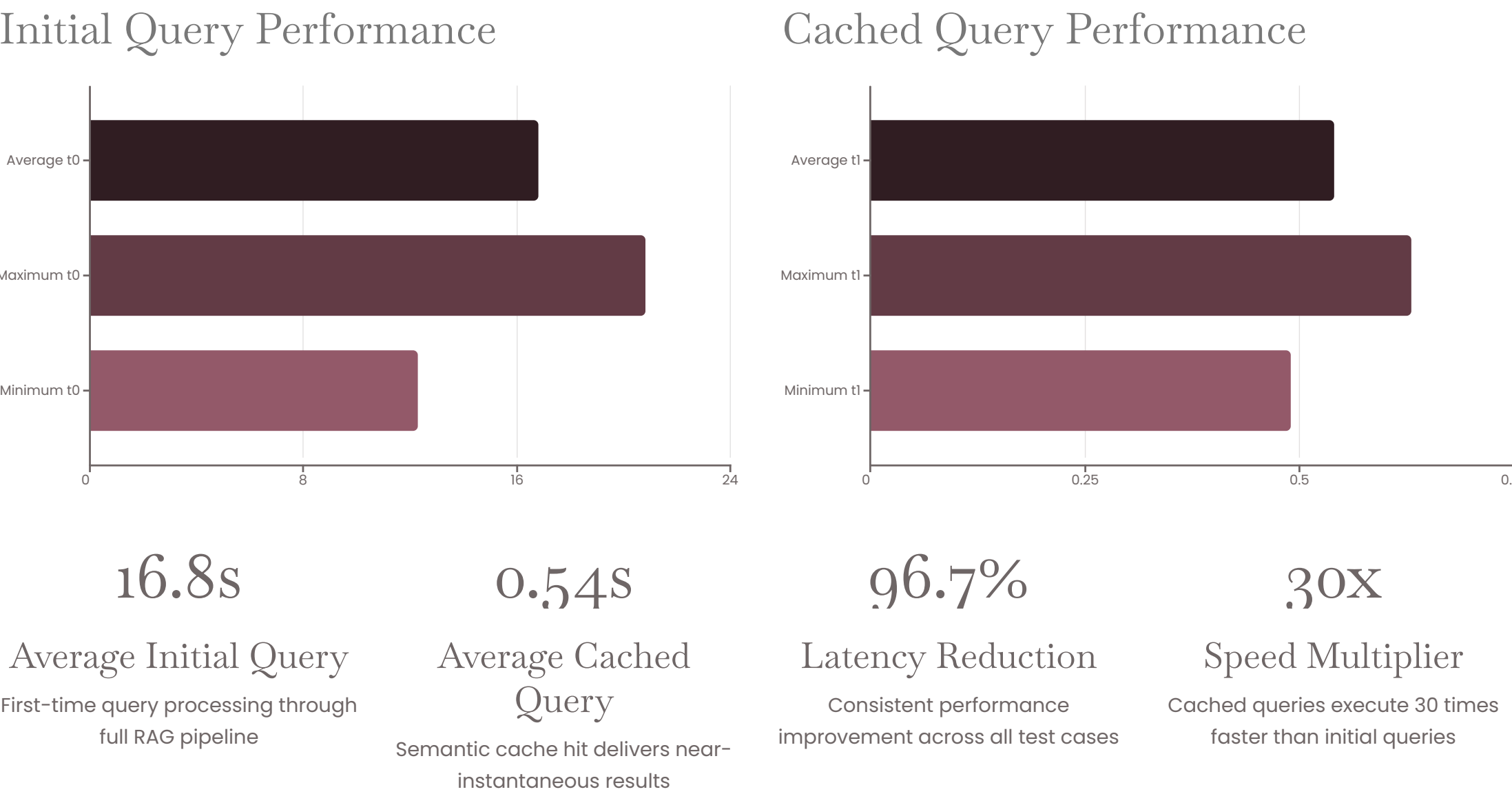
- Ollama Llama 3.2
- Nomic-embed-text
- FAISS vector store
- Semantic cache layer

Test Parameters

- 50 Q&A pairs tested
- Real-world queries
- Production conditions
- Consistent methodology

Performance Gains

- 96.7% latency reduction
- Sub-second cached queries
- Consistent improvement
- Scalable architecture



Business Impact

Semantic caching transforms user experience by delivering enterprise-grade AI responses at consumer-grade speeds. This architecture enables real-time conversational AI at scale while reducing computational costs by 97%.

Cost Optimization

Dramatic reduction in compute resources for repeated queries, lowering infrastructure costs while maintaining quality

User Experience

Sub-second response times create seamless interactions, increasing engagement and satisfaction metrics

Scalability

Cache layer enables horizontal scaling to handle enterprise workloads without linear cost increases

Technical Achievement

This implementation demonstrates production-ready semantic caching with **Llama 3.2** and **nomic-embed-text**, achieving consistent sub-second performance across 50 diverse queries. The system maintains accuracy while delivering a **30x speedup** for cached requests.